

Unit 3: Numerical problems

1. After being excited, the electron of a hydrogen atom eventually falls back to the ground state. This can take place in one jump or in a series of jumps, the electron falling into lower excited states before it end up in the ground state, that is $n=3$. Calculate the different photon energies that may be emitted as the atom returns to the ground state. Ans 12.05 eV, 1.88 eV, 10.17 eV
2. A beam of monochromatic neutrons is incident on a KCl crystal lattice spacing of $3.14 \text{ \AA}$. The first order diffraction maximum is observed when the angle ' θ ' between the incident and the atomic planes is 37° . What is the kinetic energy of the neutrons. Ans: $9.21 \times 10^{-22} \text{ J}$
3. Calculate the shortest and the longest wavelength of the Balmer series of Hydrogen. Ans: $3646 \text{ \AA}$, $6565 \text{ \AA}$
4. What are a. the energy, b. the momentum, and c. the wavelength of photon that is emitted when a Hydrogen atom undergoes a transition from the state $n=3$ to $n=1$? (The momentum of the photon is given by hf/c). Ans: 12.06 eV, $6.42 \times 10^{-27} \text{ Kg ms}^{-1}$, $1.03 \times 10^{-7} \text{ m}$.
5. The shortest wavelength of the Paschen series from Hydrogen is $8204 \text{ \AA}$. From this fact, calculate the Rydberg constant. Ans: $1.097 \times 10^7 \text{ m}^{-1}$.
6. The ground state and the first excited-state energies of potassium atoms are -4.3 eV and -2.7 eV respectively. If we use potassium vapour in the Frank-Hertz experiment at what voltages would we see drops in the plot of current versus voltage. Ans 4.8 V
7. The de-broglie wavelength of proton is 10^{-13} m . a. What is the speed of the proton ? . b. Through what potential difference must the proton be accelerated to acquire such a speed ? Ans $8.20 \times 10^4 \text{ V}$
8. An α -particle is emitted from a nucleus with an energy of 5 MeV. Calculate the wavelength of an α -particle with such energy and compare it with the size of the emitting nucleus that has a radius of $8 \times 10^{-15} \text{ m}$. Ans $6.409 \times 10^{-15} \text{ m}$.
9. A neutron spectroscopy a beam of mono energetic neutrons is obtained by reflecting reactor neutrons from a beryllium crystal. If separation between the atomic planes of the beryllium crystal is $0.732 \text{ \AA}$, what is the angle between the incident neutron beam and the atomic planes that will yield a monochromatic beam of neutrons of wavelength $0.1 \text{ \AA}$. Ans 3.92°
10. A small particle of mass 10^{-6} gm moves along the x-axis; its speed is uncertain by 10^{-6} ms^{-1} .
 - a. What is the uncertainty in the x-coordinate of the particle ? Ans: $9.1 \times 10^{-31} \text{ Kg}$
 - b. Repeat the calculation for an electron assuming that the uncertainty in its velocity is also 10^{-6} ms^{-1} . Ans: 115.768 m
11. The uncertainty in the position of a particle is equal to the de-broglie wavelength of the particle. Calculate the uncertainty in the velocity of the particle in terms of the velocity of the de-Broglie wave associated with the particle? Ans $V_{\text{wave}}/2\pi$.